

CSE 421/521 PA1 Phase 2: Alarms & Busy-Waiting

The objective of this assignment is to replace busy-waiting with a more efficient way to sleep, and to pass all the alarm tests. See below for an explanation of busy-waiting; the phase 2 recitation video explains this too:

- Your team will use the same PA1 repository from Phase 1. You will implement Phase 2 over your Phase 1 implementation.
- The goal of this phase is to:
 1. Eliminate busy-waiting. We will check for this on Autolab.
 2. Pass the following tests: `alarm-single`, `alarm-multiple`, `alarm-simultaneous`, `alarm-priority`, `alarm-zero`, and `alarm-negative` + all of Phase 1's tests
- To explain busy-waiting, one way to make a thread sleep for T ticks of time is to make it spin in a loop and yield until it is time to 'wake up'. However, this is rather inefficient as it unnecessarily consumes CPU time to check if T ticks have passed.
- In fact, there is a way (a couple of ways probably) where you could make the thread sleep for T ticks without a yield-loop. You need to implement this mechanism. **If you're not sure where to start, I strongly recommend watching the Phase 2 recitation video first.**
- You need to find and edit relevant code in `pintos/src/threads` and `pintos/src/devices`.
- If you plan to run this phase's tests with `make check`, you'll need to update the `Make.tests` file and add the alarm tests. You can find the updated `Make.tests` file on UBLearn's. The starter repository on GitHub also has these updates so you can copy the changes from there; [here's a link to the commit](#).

You can find `Make.tests` in `pintos/src/tests/threads/`

- We will be using Autolab to evaluate your work. You will submit your Phase 2 implementation the same way you did with Phase 1.

This phase is small compared to Phase 3, but this is where you start to see how different components of Pintos threading interact with each other. I encourage you to work ahead of schedule to give yourselves more room for Phase 3. If you want to run more tests with `make check`, then edit `pintos/src/tests/threads/Make.tests`